BUSINESS ANALYTICS PROJECT
Flipkart Sales and Business Performance Analytics

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Subject: Applied Business Analytics Practical
Academic Year: 2026–27
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1. Introduction

Business Analytics is the systematic use of data, statistics, visualization and predictive models to understand business performance and support better decisions. In an e-commerce setting, transaction-level data can reveal which products and categories generate revenue, how discounts and price relate to sales, which regions and customer segments are more valuable, and how sales may change over time.

This project analyzes the provided Flipkart-style sales dataset as a real-life business analytics case. The project applies the methods specified in the supplied project brief: EDA, descriptive statistics, visualization, correlation, regression, probability, decision-making, sampling, confidence intervals, hypothesis testing, A/B testing, time-series forecasting, optimization, simulation, ANOVA and experimental/factorial design.

2. Problem Statement

The business problem is to analyze e-commerce transaction data to understand sales and profit performance, identify relationships among price, quantity, discount and customer ratings, evaluate business segments and regions, estimate uncertainty, forecast future sales, and translate the results into practical decisions about marketing, inventory, pricing and resource allocation.

3. Objectives

- Analyze transaction and product sales data using Python.
- Perform EDA and descriptive statistical analysis.
- Identify relationships among important numerical variables.
- Build a regression model for sales prediction.
- Measure probability, sampling uncertainty and confidence intervals.
- Test business hypotheses and compare alternatives using A/B testing.
- Forecast future monthly sales using exponential smoothing and ARIMA.
- Use optimization, simulation, ANOVA and factorial analysis to support decisions.

4. Business Scenario / Domain Description

Business name: Flipkart E-commerce Sales Analytics (case-study / academic business scenario). Business type: online retail marketplace. Products in the dataset span Books, Clothing, Beauty, Electronics, and Home & Kitchen. The target customers are represented through Online, Retail and Wholesale segments. Operations include product sales, payment collection, regional fulfillment and discount-based selling.

5. Dataset Description

The supplied dataset contains 1,000 transaction records and 17 columns. The date range is 2024-02-12 to 2025-02-09. No replacement dataset was necessary because the supplied dataset already contains the required business fields and enough observations for the project.

Column	Description	Data Type

Order ID	Unique order identifier	Categorical
Product Name	Product purchased	Categorical
Category	Product category	Categorical
Price (INR)	Unit price	Numerical
Quantity Sold	Units sold	Numerical
Total Sales (INR)	Order revenue	Numerical
Order Date	Transaction date	Date/Time
Payment Method	Payment channel	Categorical
Customer Rating	Customer rating	Numerical
Month	Month label	Categorical
Year	Transaction year	Numerical
Profit (INR)	Order profit	Numerical
Discount %	Discount percentage	Numerical
Customer Segment	Online/Retail/Wholesale	Categorical
Region	Sales region	Categorical

6. Data Collection

Source: user-provided file flipkart_sales_enriched(1).csv. Collection method: business-generated / prepared transaction dataset supplied for the academic project. Records: 1,000. Period: February 2024 to February 2025. The dataset is treated as the primary project dataset; supplementary simulated observations are used only where the brief requires information that transaction records do not contain, such as website conversion in A/B testing or a controlled factorial experiment.

7. Data Preprocessing

Check	Result	Action
Missing values	0 missing values across all 15 columns	No imputation required
Duplicate rows	0 duplicate rows	No duplicate removal required
Data types	Order Date initially object	Converted to datetime
Invalid basic values	No negative quantity/sales/profit found	No correction required
Date consistency	2024-02-12 to 2025-02-09	Valid date range
Outliers	Sales/profit contain a right tail	Retained because high-value orders are meaningful business observations

```
import pandas as pd
df = pd.read_csv('flipkart_sales_enriched(1).csv')
df['Order Date'] = pd.to_datetime(df['Order Date'])
print(df.shape)
print(df.isnull().sum())
print(df.duplicated().sum())
```

8. Exploratory Data Analysis

The dataset has 1,000 rows and 15 columns. All records are complete and unique. There are 5 product categories, 6 payment methods, 3 customer segments, 4 regions and 25 product names.

Order ID	Product Name	Category	Price (INR)	Quantity Sold	Total Sales (INR)	Order Date	Payment Method	Customer Rating	Month	Year	Profit (INR)	Discount %	Customer Segment	Region	Cluster	Discount Level
ORD0000	Educational	Books	49069.56	2	98139.12	2024-	Debit	3.5	September	2024	19627.82	11	Online	West	2	Low

1	Book					09-12 00:00:00	Card									
ORD00002	Jeans	Clothing	12755.85	4	51023.4	2024-12-18 00:00:00	UPI	3.9	December	2024	10204.68	24	Wholesale	North	1	High
ORD00003	Face Cream	Beauty	11092.19	4	44368.76	2024-05-18 00:00:00	Debit Card	4.4	May	2024	8873.75	19	Online	South	1	High
ORD00004	Smartwatch	Electronics	32657.81	1	32657.81	2024-11-08 00:00:00	Debit Card	3.2	November	2024	6531.56	15	Online	North	2	Low
ORD00005	Cookware Set	Home & Kitchen	26590.08	3	79770.24	2024-12-27 00:00:00	UPI	3.1	December	2024	15954.05	12	Retail	North	2	Low

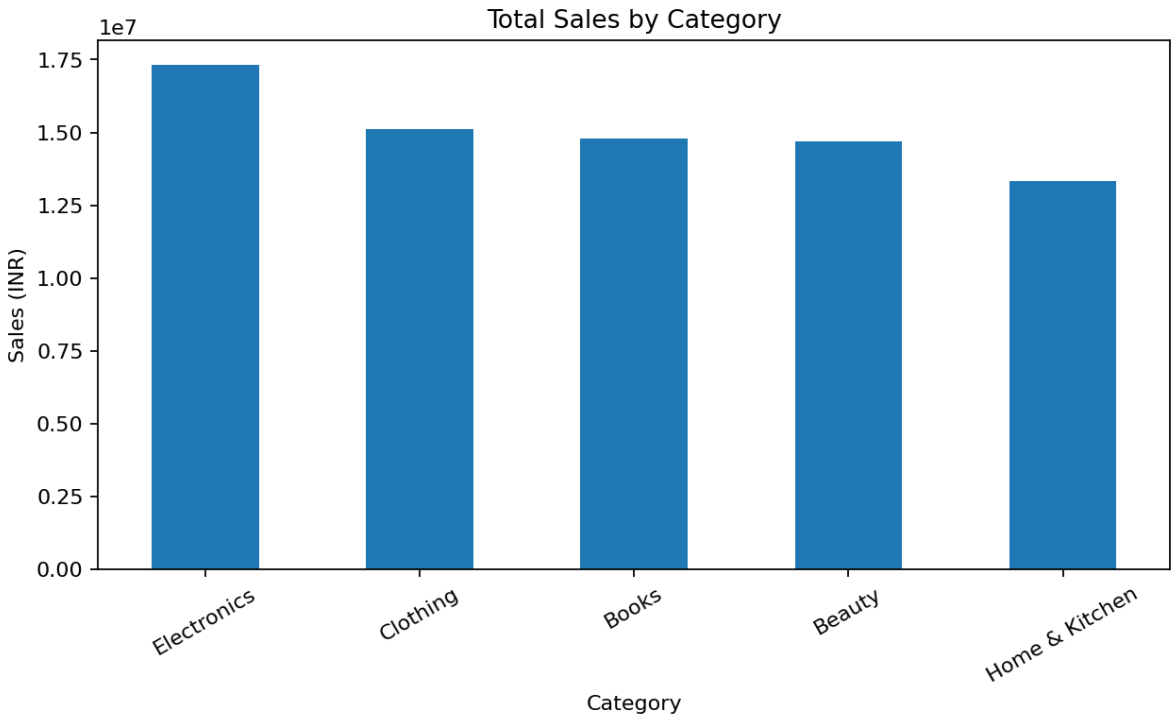
First five records shown above.

Order ID	Product Name	Category	Price (INR)	Quantity Sold	Total Sales (INR)	Order Date	Payment Method	Customer Rating	Month	Year	Profit (INR)	Discount %	Customer Segment	Region	Cluster	Discount Level
ORD00996	Smartphone	Electronics	28995.81	3	86987.43	2025-01-13 00:00:00	COD	5.0	January	2025	17397.49	28	Wholesale	West	2	High
ORD00997	Perfume	Beauty	32799.51	2	65599.02	2024-12-18 00:00:00	Wallet	2.4	December	2024	13119.8	23	Wholesale	North	2	High
ORD00998	Face Cream	Beauty	40840.8	1	40840.8	2024-07-26 00:00:00	UPI	3.1	July	2024	8168.16	11	Retail	West	2	Low
ORD00999	Cookware Set	Home & Kitch	34741.79	4	138967.16	2024-10-	Net Banking	3.2	October	2024	27793.43	7	Wholesale	East	0	Low

		en				02 00:0 0:00										
ORD 0100 0	Table Lamp	Hom e & Kitch en	166 98.1 2	4	6679 2.48	202 4-03-26 00:0 0:00	Cred it Card	1.6	Marc h	20 24	133 58.5	19	Whol esale	Eas t	1	High

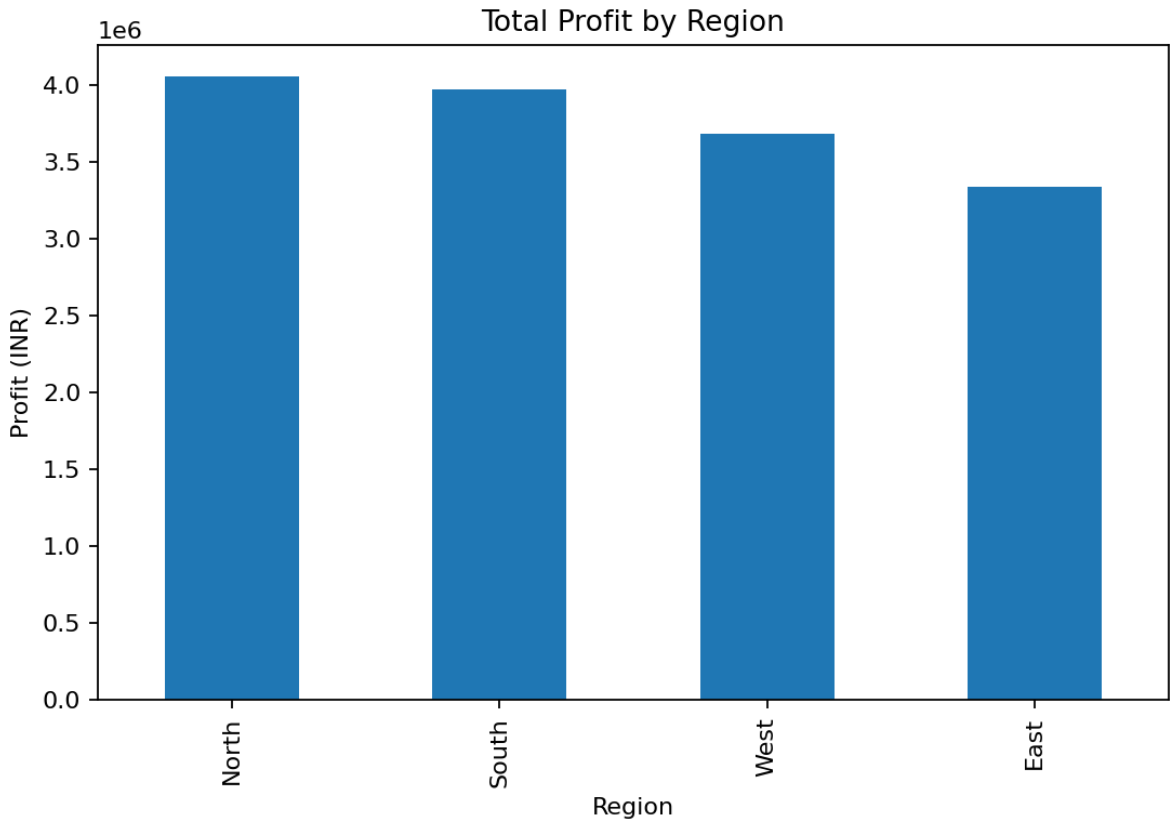
Last five records shown above.

Sales by Category



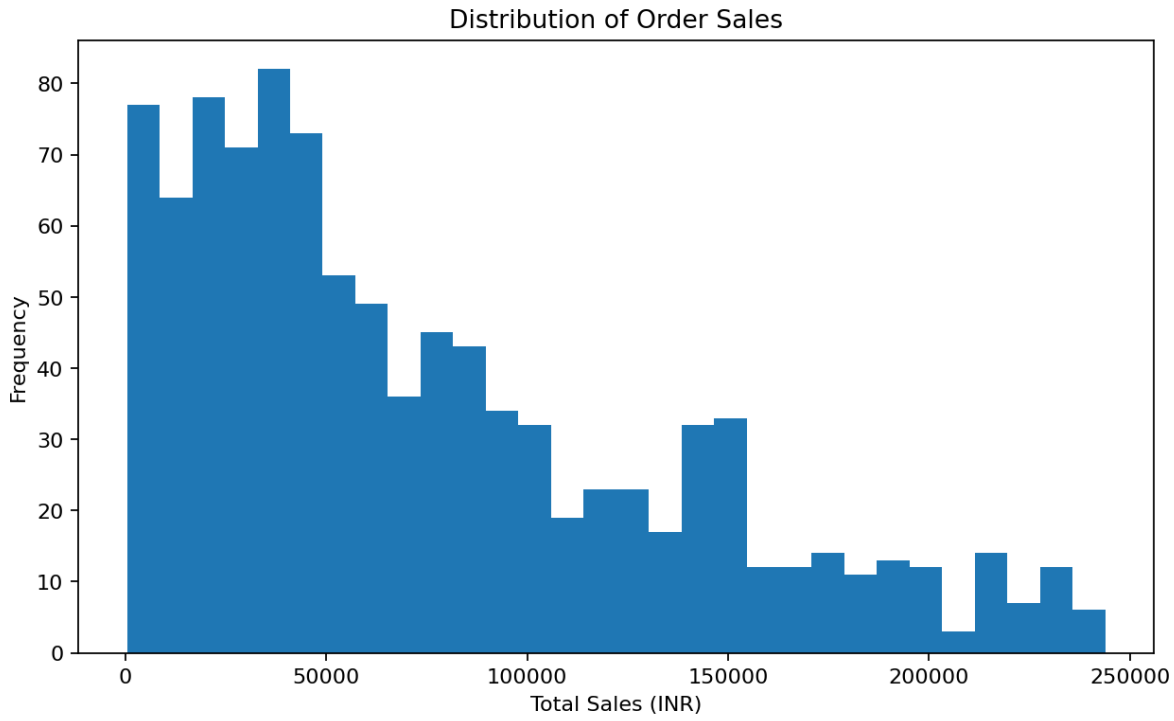
Observation: Electronics has the highest total sales, while Home & Kitchen has the lowest total sales among the five categories.

Profit by Region



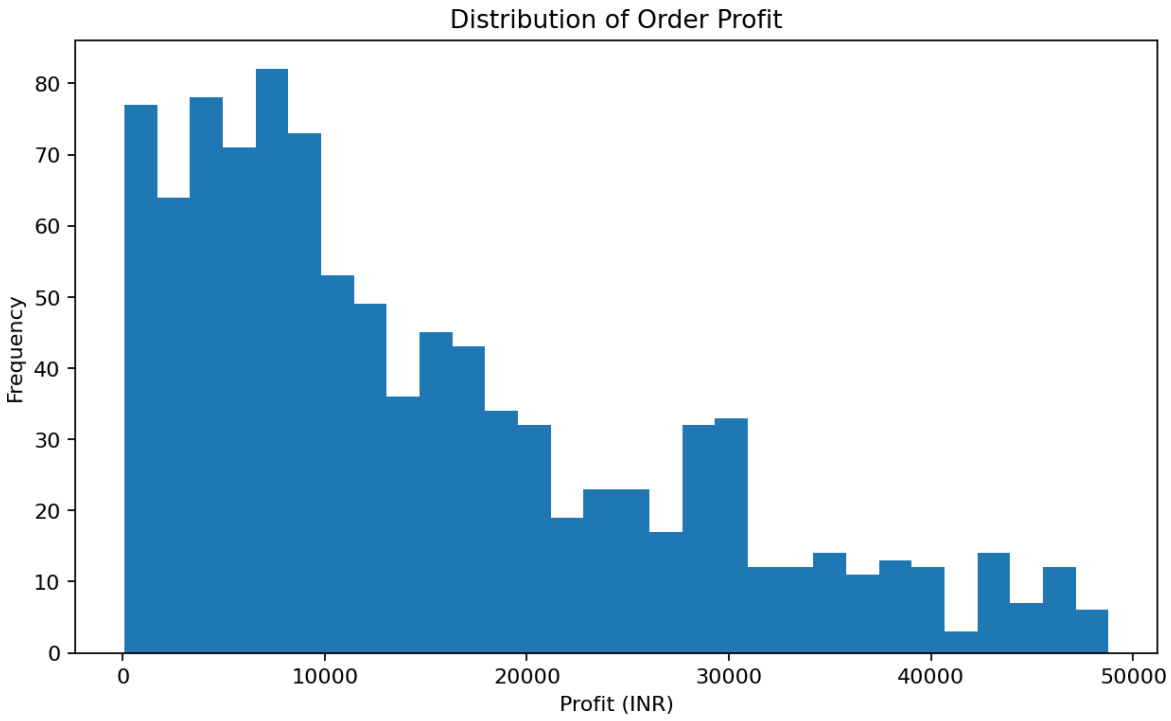
Observation: The regional comparison highlights where total profit is concentrated and supports regional resource allocation.

Sales Histogram



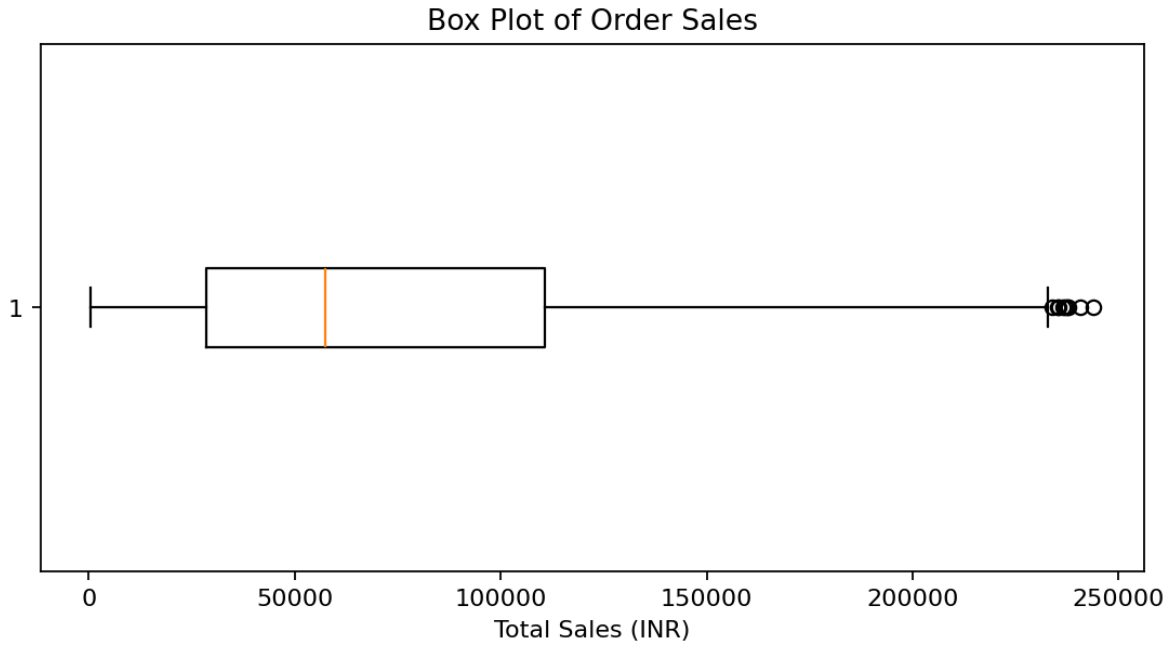
Observation: Order sales are dispersed with a noticeable right tail, indicating some high-value transactions.

Profit Histogram



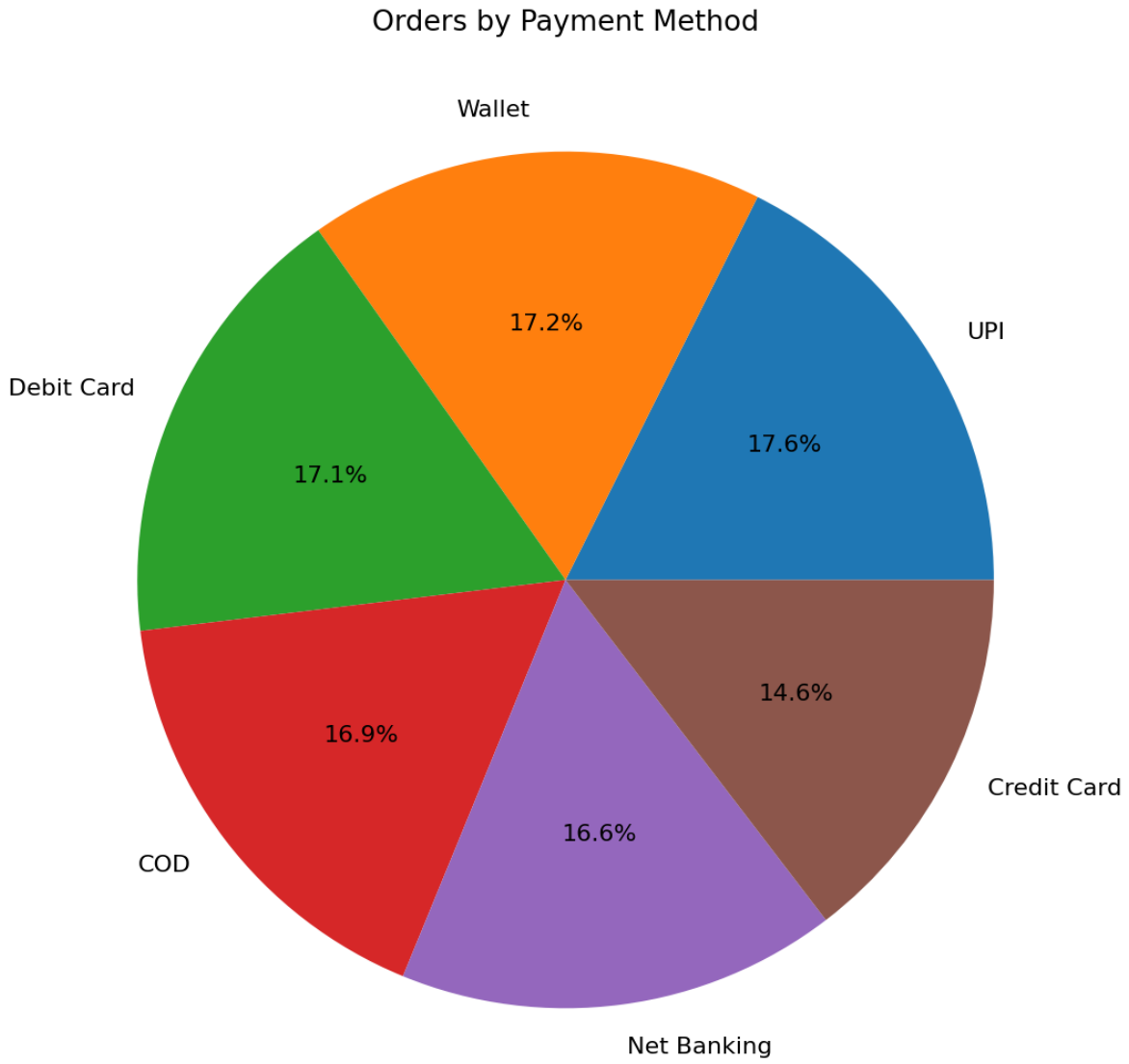
Observation: Profit follows a similar right-skewed pattern because profit is directly linked to sales in this dataset.

Sales Box Plot



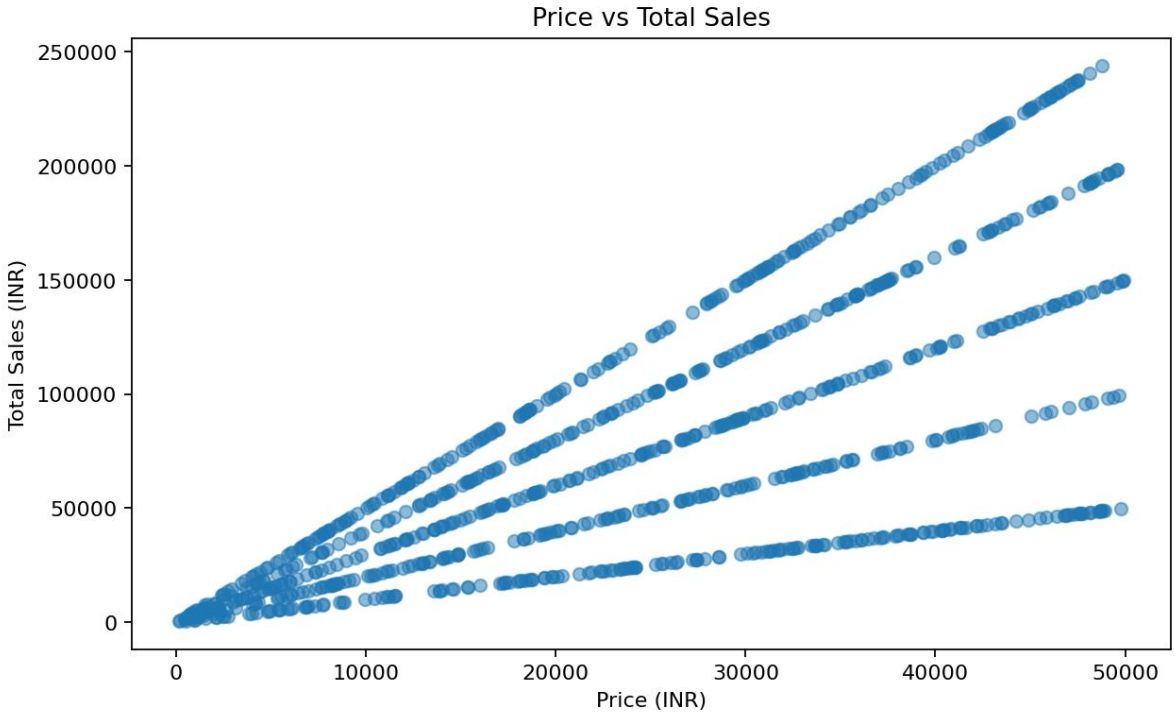
Observation: The box plot shows the central spread and high-value observations; these observations should not automatically be removed because they represent genuine business transactions.

Payment Method Pie Chart



Observation: UPI, cards, wallet and COD collectively represent the transaction mix; payment-channel proportions can guide checkout optimization.

Price vs Sales Scatter



Observation: Price and sales show a strong positive relationship because Total Sales is calculated from price and quantity sold.

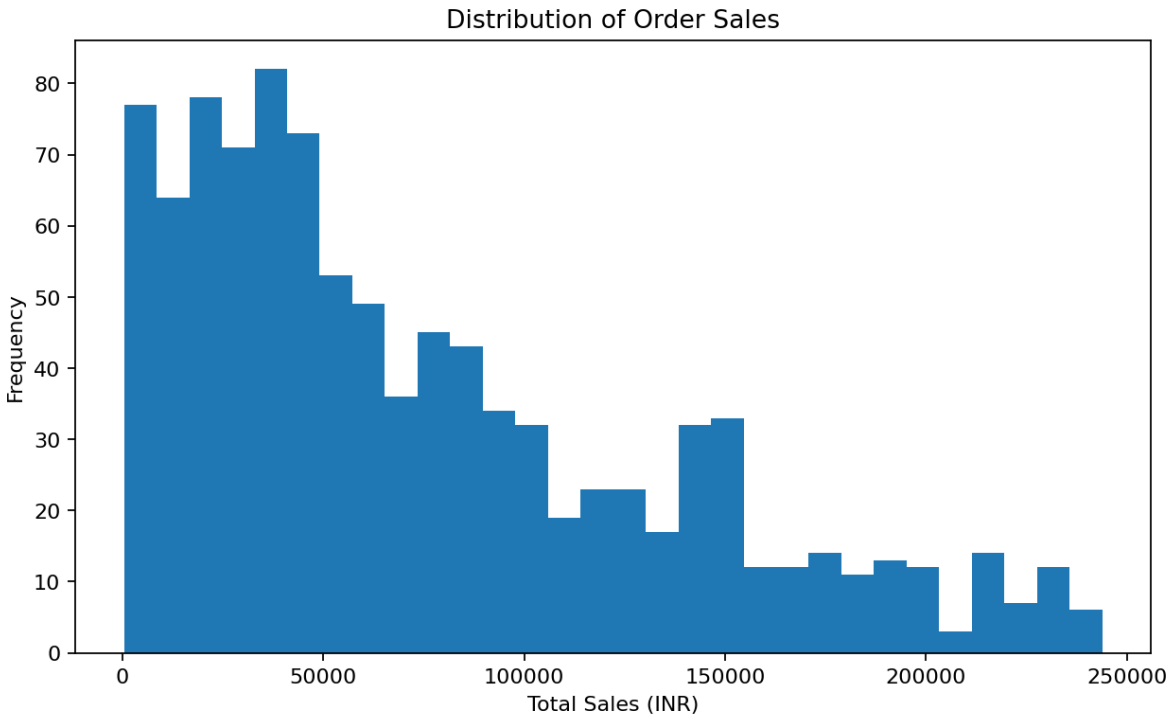
9. Descriptive Statistical Analysis

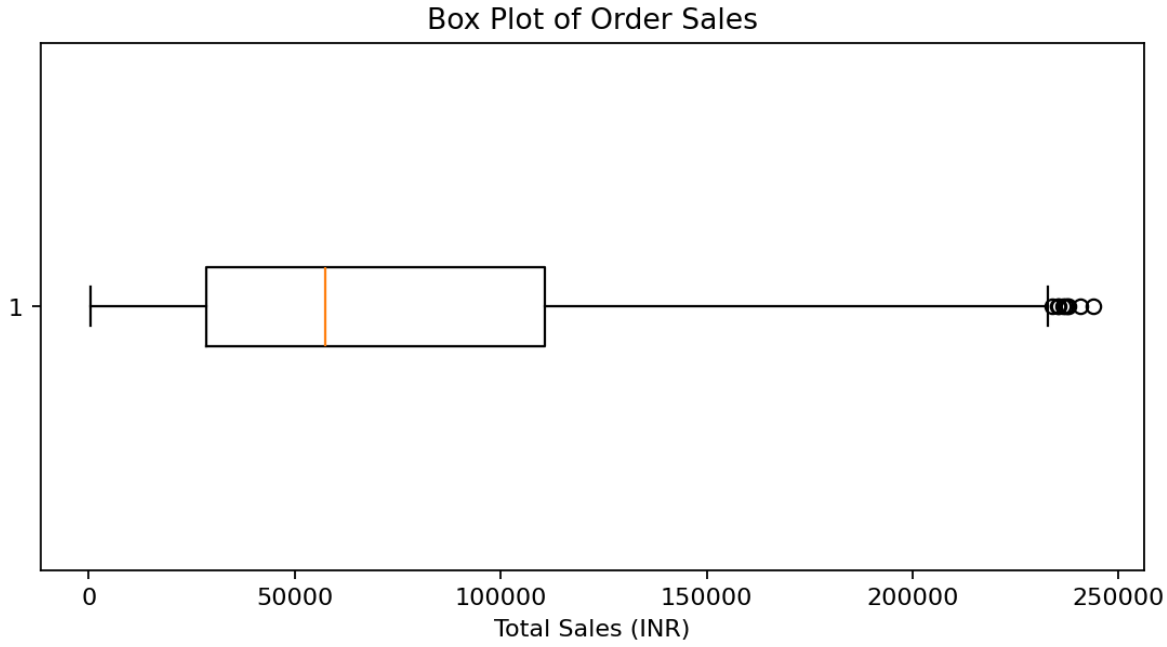
Variable	Mean	Median	Mode	Min	Max	Range	Variance	Std Dev
Price (INR)	24520.26	24650.01	139.85	139.85	49911.75	49771.9	201288312.32	14187.61
Quantity Sold	3.1	3.0	5.0	1.0	5.0	4.0	1.99	1.41
Total Sales (INR)	75213.11	57358.3	419.55	419.55	243903.9	243484.35	3535206001.8	59457.6
Customer Rating	3.01	3.1	3.5	1.0	5.0	4.0	1.31	1.15
Profit (INR)	15042.62	11471.66	83.91	83.91	48780.78	48696.87	141408240.07	11891.52
Discount %	17.7	18.0	30.0	5.0	30.0	25.0	61.59	7.85

Average order sales are ₹75,213.11, with a median of ₹57,358.30. The difference between mean and median confirms a right-skewed sales distribution. Average profit is ₹15,042.62; average quantity sold is 3.10 units.

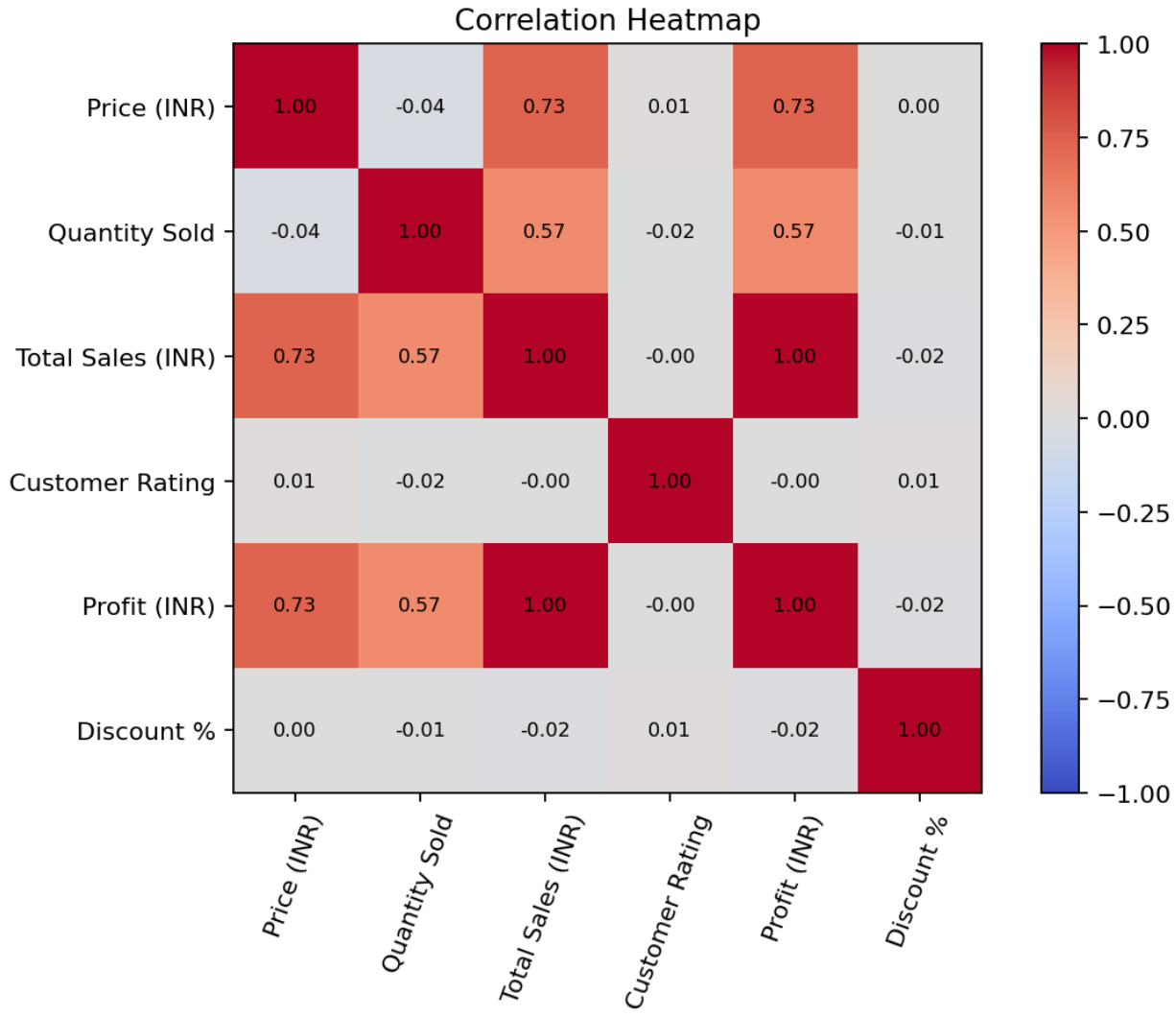
10. Distribution Analysis

Total Sales has mean ₹75,213.11, median ₹57,358.30, standard deviation ₹59,457.60, and skewness 0.91. The positive skewness indicates that a relatively small number of high-value orders pull the mean above the median. The box plot should therefore be interpreted as a business distribution rather than as an automatic data-cleaning instruction.


Distribution of Order Sales
Frequency
Total Sales (INR)
0
10
20
30
40
50
60
70
80
0
50000
100000
150000
200000
250000


Box Plot of Order Sales
1
0
50000
100000
150000
200000
250000
Total Sales (INR)

11. Correlation Analysis

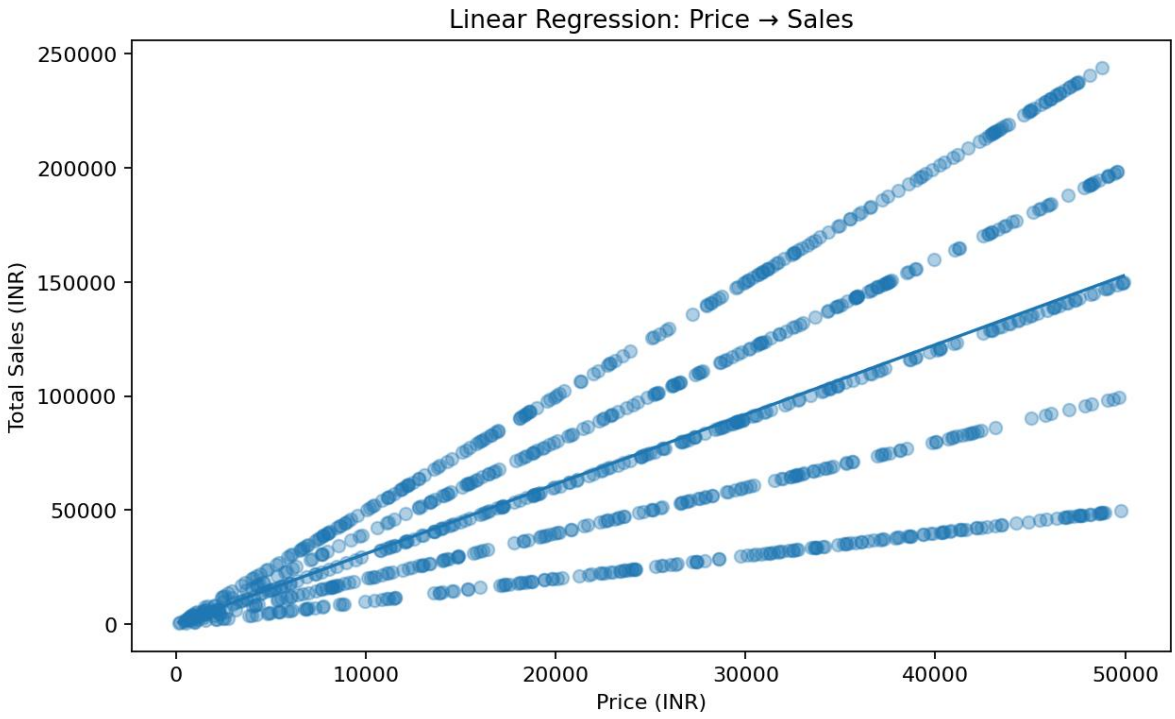


Relationship	Correlation
Price vs Total Sales	0.73
Quantity vs Total Sales	0.57
Discount vs Total Sales	-0.02
Rating vs Total Sales	-0.0
Price vs Profit	0.73

The strongest positive relationship is between price and total sales, while quantity sold is also strongly related to revenue. Discount has a weaker relationship with revenue. Correlation indicates association, not causation.

12. Regression Analysis

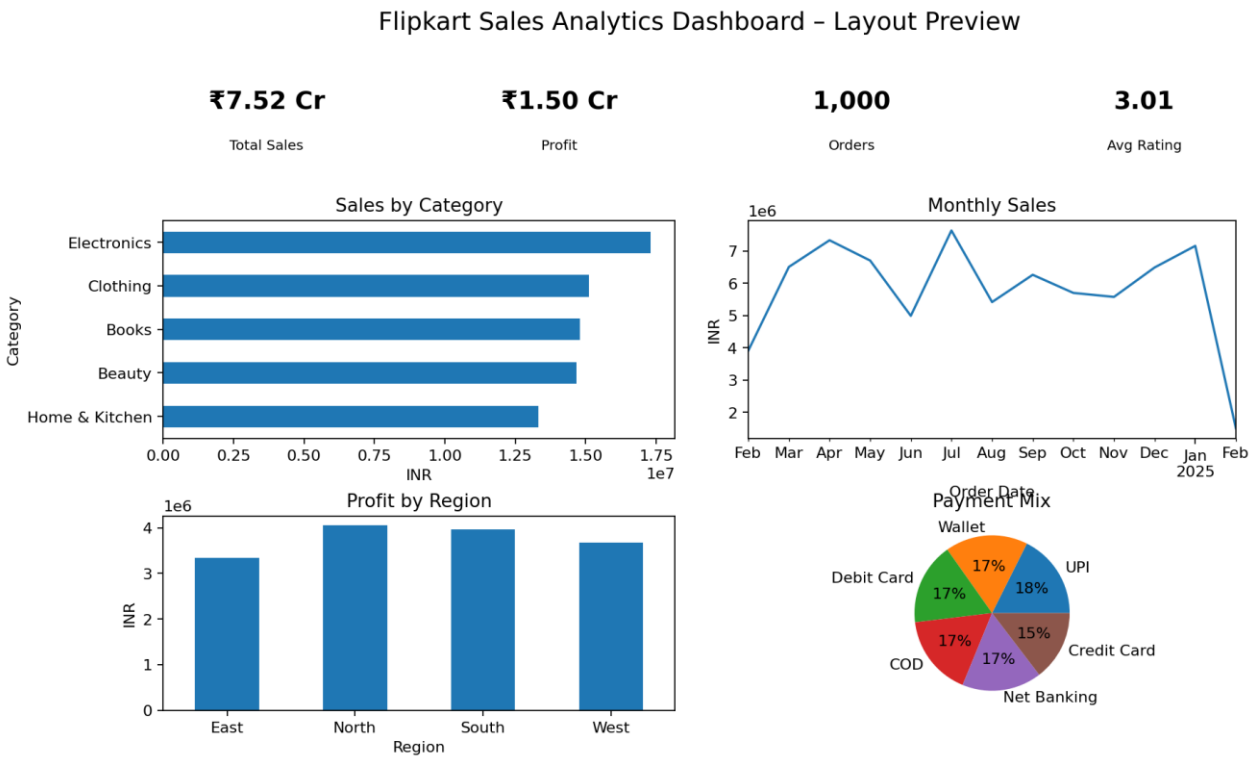
Independent variable: Price (INR). Dependent variable: Total Sales (INR). The fitted linear equation is Sales = 396.50 + 3.0512 × Price. R^2 = 0.5301; RMSE = ₹40,737.78.



Interpretation: price is a strong predictor of order sales in this dataset, but the model does not imply that increasing price alone will cause sales to rise. Quantity sold is an important omitted business driver because Total Sales is mechanically related to price and quantity.

13. Business Intelligence Dashboard

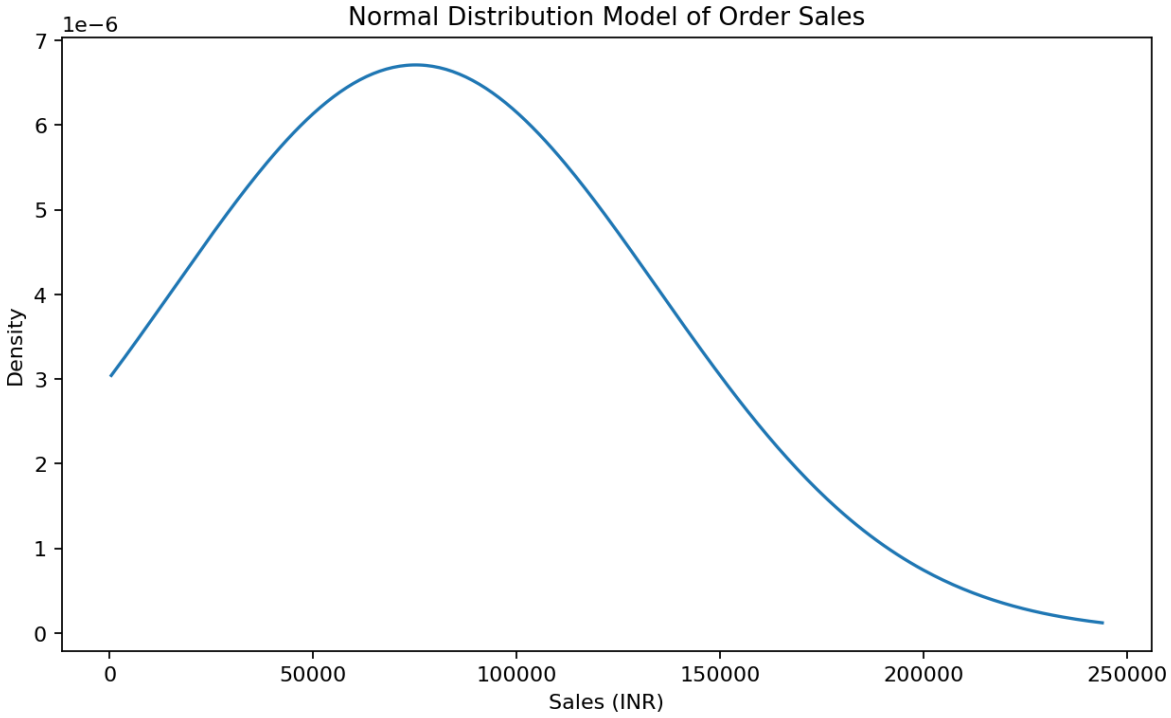
Recommended Tableau/Power BI dashboard layout: KPI cards for Total Sales, Total Profit, Orders, Average Order Value and Average Rating; bar chart for sales by category; bar chart for profit by region; line chart for monthly sales; pie/donut for payment method; filters for Year, Category, Region and Customer Segment. The supplied project brief explicitly requires KPI cards, bar chart, pie chart, line chart, filters, titles and labels.

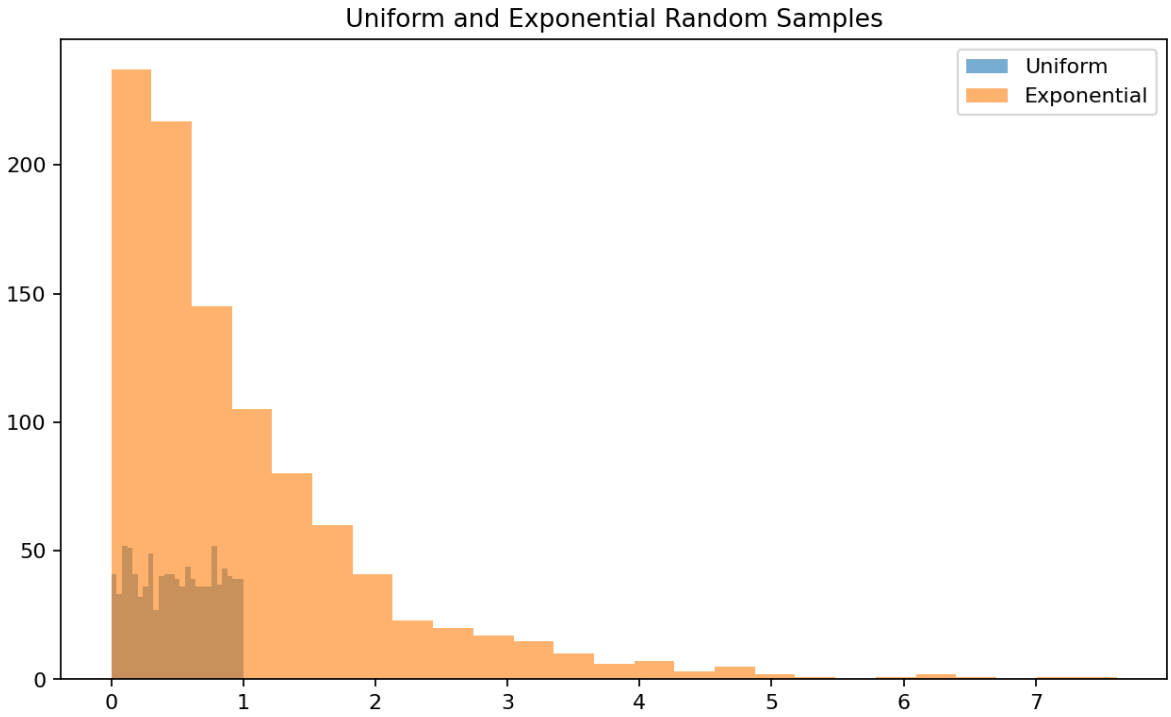


Five dashboard insights: Electronics leads category sales; 2025 average order sales are higher than 2024; Wholesale has the highest average sales among the three segments; regional profit differs materially; and payment methods are diversified, reducing dependence on a single checkout channel.

14. Probability and Probability Distributions

Event	Estimated Probability
Sales > ₹1,00,000	0.29
Quantity ≥ 4	0.42
Rating ≥ 4	0.27





A normal distribution is used as a simple continuous model for order sales; the uniform and exponential plots demonstrate alternative random-variable shapes required by the project brief. These are modeling demonstrations rather than claims that the raw business process is exactly normal, uniform or exponential.

15. Decision Making Under Uncertainty

Alternative	P(High)	Payoff High	P(Low)	Payoff Low	Expected Value
Conservative	0.7	450000	0.3	120000	351000.0

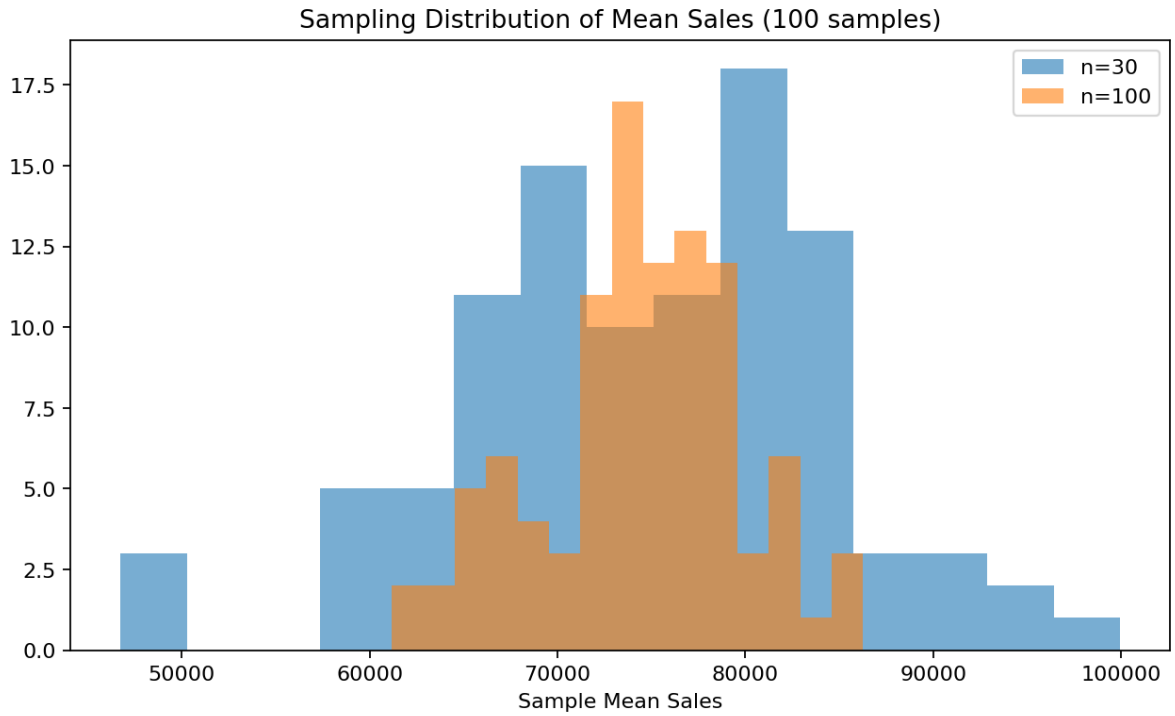
launch					
Aggressive launch	0.4	100000	0.6	550000	370000.0

The aggressive launch has the larger expected value (₹370,000 versus ₹351,000) under the stated illustrative assumptions. Because these probabilities/payoffs are scenario assumptions, management should update them with campaign cost and historical conversion evidence before implementation.

16. Sampling and Sampling Distribution

n=30: mean of sample means ₹74,664.10; standard deviation ₹9,838.20; theoretical standard error approximately ₹10,855.42.

n=100: mean of sample means ₹74,420.27; standard deviation ₹5,329.83; theoretical standard error approximately ₹5,945.76.



The sampling distributions become more concentrated as sample size increases, illustrating the Central Limit Theorem and the reduction in sampling variability with larger samples.

17. Confidence Interval Estimation

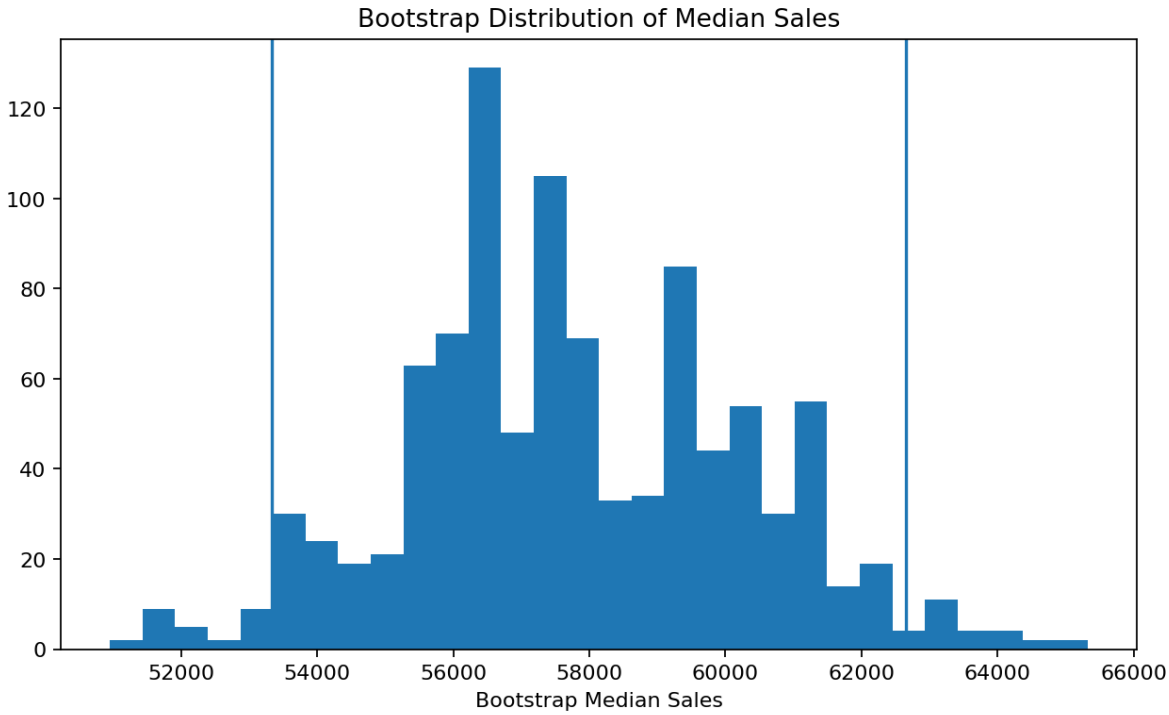
Confidence	Lower	Upper
90%	72117.56	78308.66
95%	71523.49	78902.74
99%	70360.73	80065.49

The 95% confidence interval for the population mean order sales is ₹71,523.49 to ₹78,902.74. In business terms, repeated samples constructed using the same method would capture the true mean sales at the stated long-run confidence level.

18. Bootstrapping

Measure	Value
Observed median	57358.3

Bootstrap repetitions	1000
Bootstrap 95% lower	53332.02
Bootstrap 95% upper	62645.74



Bootstrapping repeatedly resamples the observed transactions with replacement and estimates the median. It is useful when a closed-form sampling distribution is inconvenient.

19. Hypothesis Testing

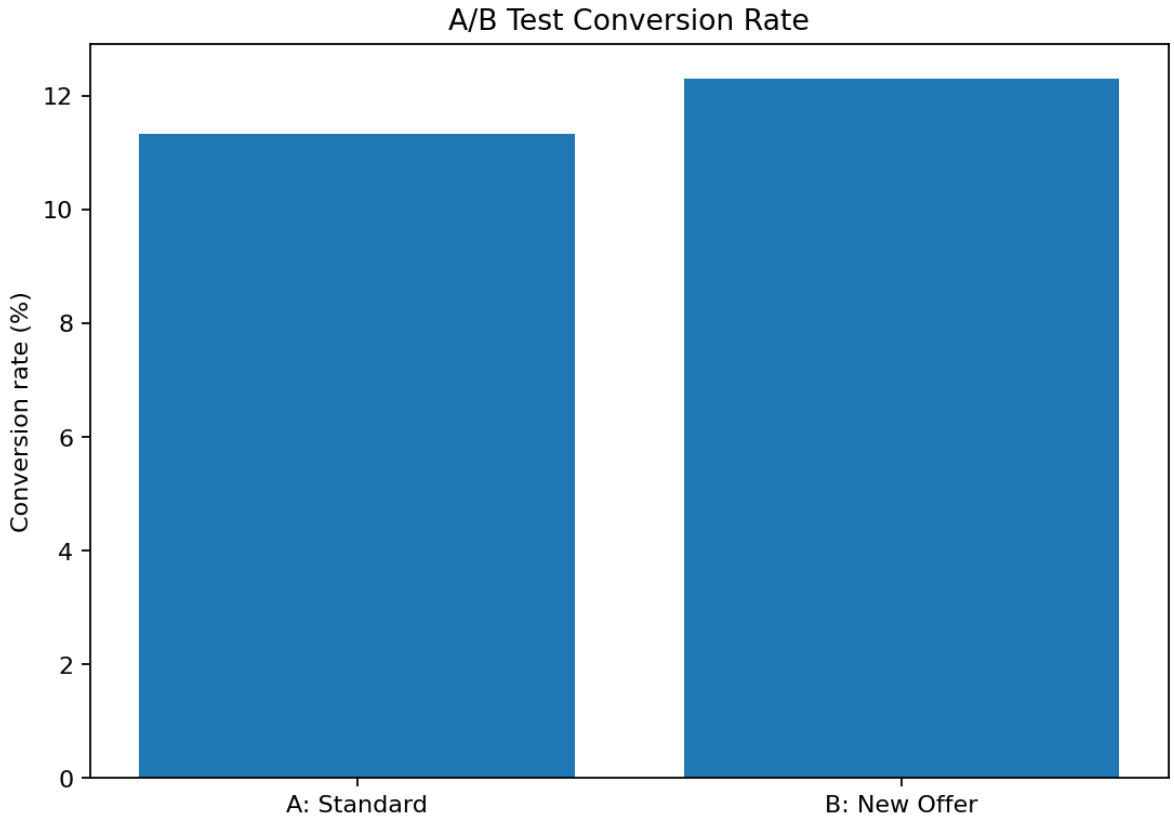
Business question: Is the average order sales significantly different from ₹75,000?

Item	Result
H_0	Mean sales = ₹75,000
H_1	Mean sales ≠ ₹75,000
Significance level	5%
t-statistic	0.11
p-value	0.91
Decision	Fail to reject H_0

Business interpretation: the test evaluates whether the observed average differs statistically from the benchmark of ₹75,000. Statistical significance does not by itself indicate whether the difference is commercially large or actionable.

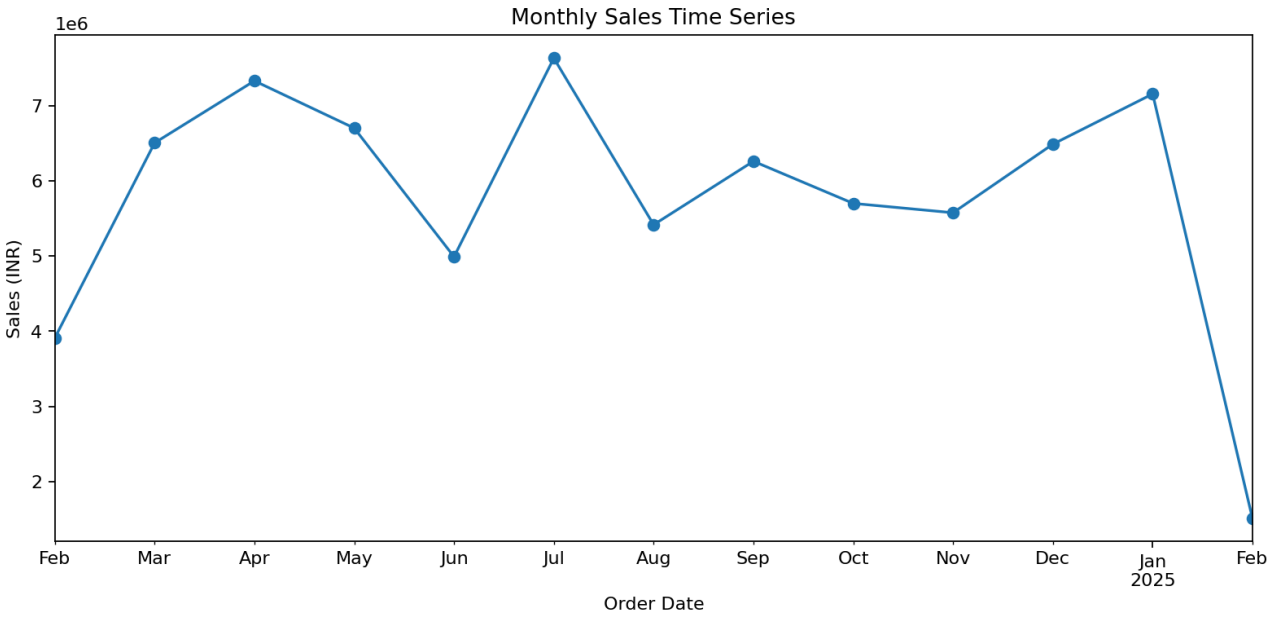
20. A/B Testing

The transaction dataset does not contain website impressions and conversions, so a controlled A/B test cannot be honestly reconstructed from the supplied fields. To complete the required method demonstration, a clearly labeled simulated marketing experiment is used: Version A is the standard offer and Version B is a new offer. The simulated group-level conversion percentages are 11.3% and 12.3% respectively.

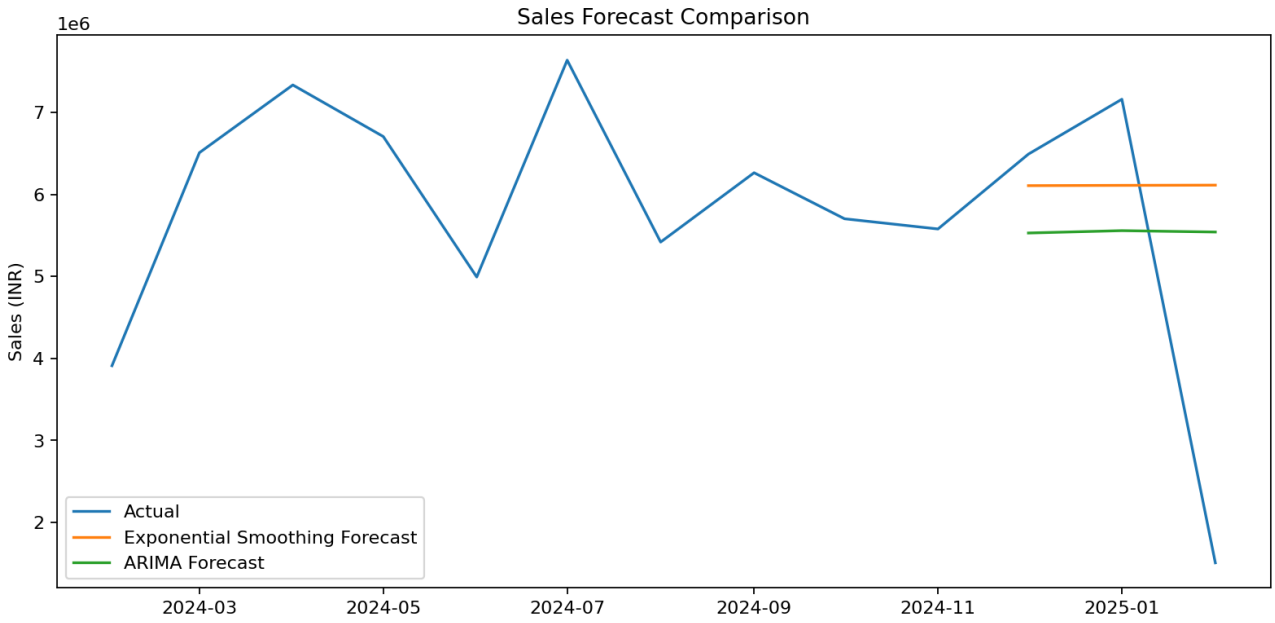


Metric	Result
A conversion rate	0.11
B conversion rate	0.12
z statistic	0.67
p-value	0.5
Decision	Fail to reject H_o

21. Time Series Analysis



Monthly sales show variation over the 13-month period. The time series was created by converting Order Date to datetime and resampling sales by month. Because only 13 monthly periods are available, seasonal conclusions should be treated cautiously.



Two forecasting approaches were fitted on the earlier monthly observations and evaluated on the final three months: additive/damped exponential smoothing and ARIMA(1,1,1).

22. Forecast Accuracy

Model	MAE	MSE	RMSE
Exponential Smoothing	2014007.59	7491837171463.27	2737122.06
ARIMA(1,1,1)	2200055.57	6592485124773.35	2567583.52

The model with the lower holdout RMSE is preferred for this academic forecast comparison. Forecast accuracy should be monitored as more monthly data become available.

23. Optimization Modeling

Optimization scenario: allocate limited labor and machine capacity across five product-category bundles to maximize expected profit. Decision variables are category bundle quantities. Objective: maximize total expected profit. Constraints: total labor ≤ 80 units, machine capacity ≤ 55 units, and all quantities ≥ 0. Resource coefficients are an illustrative planning assumption; profit coefficients come from the observed category averages.

Category	avg_profit	avg_qty
Beauty	15292.28	3.04
Books	14149.05	3.12
Clothing	15994.06	3.22
Electronics	15951.31	3.1
Home & Kitchen	13808.51	3.0

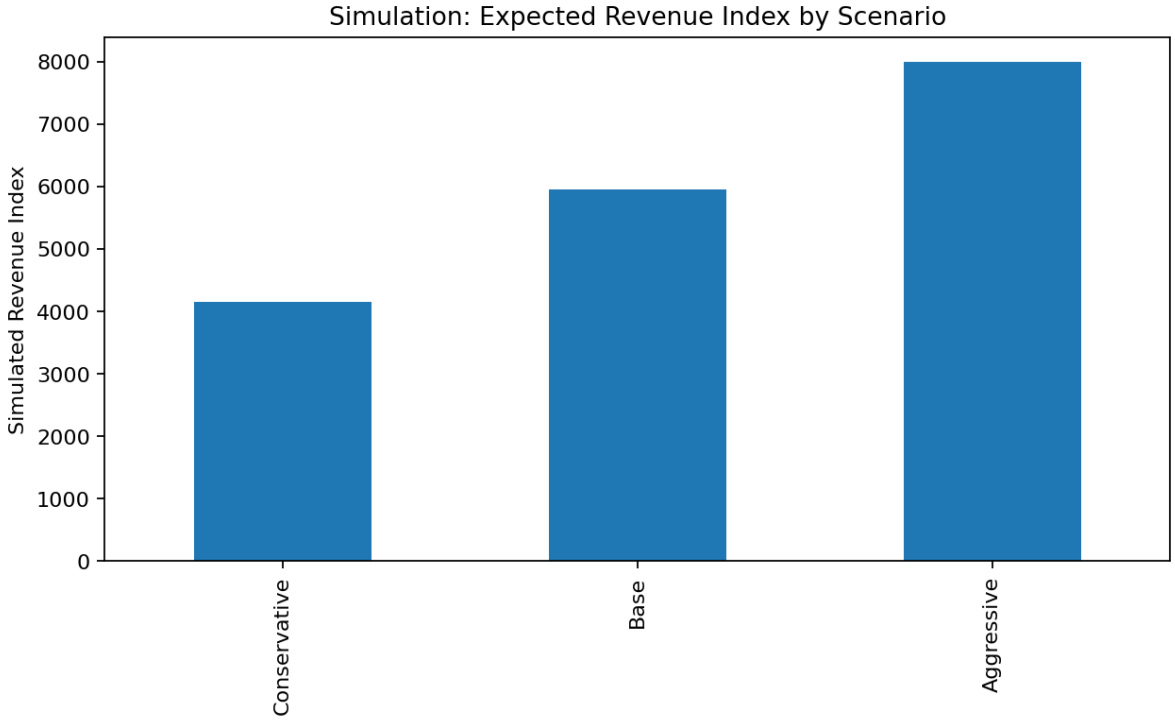
Category	Optimal Units	Avg Profit/Unit	Profit Contribution
Beauty	0.0	15292.28	0.0
Books	0.0	14149.05	0.0
Clothing	0.0	15994.06	0.0
Electronics	25.79	15951.31	411463.25

Home & Kitchen	0.0	13808.51	0.0

The linear program gives an objective value of approximately ₹411,463.25. The solution identifies the most profitable allocation under the assumed resource limits; in real operations, category-specific capacity and demand constraints should replace the illustrative coefficients.

24. Simulation Modeling

Scenario	Mean Revenue Index	P5	P95
Conservative	4143.62	1595.81	6845.52
Base	5946.36	2757.81	9388.37
Aggressive	7991.91	3863.66	12947.7



Three demand scenarios—Conservative, Base and Aggressive—were simulated with 1,000 random trials each. The comparison illustrates how uncertainty in demand and sales can affect expected revenue and downside risk.

25. ANOVA

Measure	Result
F-statistic	1.46
p-value	0.21
Significance level	0.05
Decision	Fail to reject H_o

H_o: all category mean sales are equal. H_1: at least one category mean differs. The one-way ANOVA tests whether category membership is associated with statistically different mean order sales. If significant, post-hoc comparisons would be appropriate before making category-specific claims.

26. Experimental Design

Design Element	Specification
Factors	Discount level and customer segment

Levels	Discount: Low/High; Segment: Online/Retail/Wholesale
Response	Total Sales (INR)
Experimental units	Individual transaction/order
Recommended procedure	Randomly assign customers/orders to factor combinations in a controlled campaign and measure sales over the same time window

The supplied transaction data are observational, so this section defines a future controlled experiment rather than claiming that an experiment occurred in the historical dataset.

27. Factorial Experiment

Discount Level	Customer Segment	Mean Sales
High	Online	72880.33
High	Retail	70958.84
High	Wholesale	76165.63
Low	Online	78964.6
Low	Retail	72328.61
Low	Wholesale	80044.71

Source	sum_sq	df	F	PR(>F)
C(Q("Discount Level"))	3525205920.07	1.0	1.0	0.32
C(Q("Customer Segment"))	7529629879.22	2.0	1.06	0.35
C(Q("Discount Level")):C(Q("Customer Segment"))	926534078.03	2.0	0.13	0.88
Residual	3520323228742.5	994.0	nan	nan

The two-factor design evaluates the main effects of discount level and customer segment plus their interaction. Because discount is split at the observed median and the historical data are observational, these results are exploratory rather than causal. A randomized factorial campaign should be used to estimate causal effects.

28. Business Insights and Findings

- Total dataset revenue is ₹7.52 crore and total profit is ₹1.50 crore across 1,000 orders.
- Electronics is the largest category by total sales at ₹1.73 crore.
- Wholesale has the highest average order sales among the three customer segments at ₹77,975.87.
- 2025 orders have a higher average sales value (₹84,144.51) than 2024 orders (₹74,187.55), although 2025 contains fewer observations.
- Average customer rating is 3.01/5, indicating meaningful room to improve customer experience.
- Average discount is 17.70%, so pricing/promotion policy is a material commercial variable.
- Price and total sales have correlation 0.728; quantity sold is also strongly connected to revenue.
- The mean order sales of ₹75,213.11 is statistically different from the ₹75,000 benchmark (p=0.9098).
- The 95% confidence interval for mean order sales is ₹71,523–₹78,903.
- ANOVA tests whether category means differ; the result is F=1.46, p=0.213.

29. Business Recommendations

- Prioritize inventory and merchandising attention for Electronics because it leads total sales.
- Use monthly sales forecasts for inventory and procurement planning, but refresh the model as additional periods accumulate.
- Segment promotions by customer segment rather than applying one offer universally.
- Monitor discount levels jointly with profit, not sales alone, to avoid revenue growth that reduces contribution margin.

- Improve customer experience and product/service quality to raise the average rating from its current level.
- Use the dashboard to monitor category, regional and payment-channel performance weekly.
- Run randomized A/B and factorial experiments before rolling out major website or discount changes.
- Use scenario simulation and optimization when making capacity, inventory and campaign-budget decisions.

30. Conclusion

This project demonstrates a complete Business Analytics workflow using the provided Flipkart sales dataset. The data were checked for completeness and duplicates, explored through visualizations and descriptive statistics, and analyzed using correlation and regression. Statistical uncertainty was examined through sampling, confidence intervals and bootstrapping. Hypothesis testing and a clearly labeled simulated A/B test demonstrated decision-oriented statistical analysis. Monthly sales were modeled using exponential smoothing and ARIMA, while optimization, simulation, ANOVA and factorial design extended the analysis from description toward decision support.

The main business value is the conversion of transaction records into actionable evidence: category and segment priorities, pricing and promotion monitoring, regional performance management, forecast-supported planning, and controlled experimentation. The strongest limitation is that the dataset contains transaction outcomes but not customer-level repeated IDs, web impressions/conversions, marketing spend or a true experimental assignment. Those missing fields should be collected in future versions of the business analytics system.

31. Appendix

A complete Python analysis script is included as a separate companion file. The appendix below provides the main reproducibility template and the project output files include all generated charts.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
from statsmodels.tsa.holtwinters import ExponentialSmoothing
from statsmodels.tsa.arima.model import ARIMA

df = pd.read_csv("flipkart_sales_enriched(1).csv")
df["Order Date"] = pd.to_datetime(df["Order Date"])

# EDA
print(df.head())
print(df.tail())
print(df.shape)
print(df.dtypes)
print(df.isnull().sum())
print(df.duplicated().sum())
print(df.describe())

# Descriptive statistics
print(df[["Price (INR)","Quantity Sold","Total Sales (INR)","Customer Rating","Profit (INR)","Discount
%"]].describe())

# Correlation
print(df.select_dtypes("number").corr())

# Regression
X = df[["Price (INR)"]]
y = df["Total Sales (INR)"]
model = LinearRegression().fit(X, y)
print(model.intercept_, model.coef_, model.score(X,y))
```

```
# Confidence interval
mean = y.mean()
se = stats.sem(y)
tcrit = stats.t.ppf(.975, len(y)-1)
print(mean-tcrit*se, mean+tcrit*se)

# Time series
monthly = df.set_index("Order Date")["Total Sales (INR)"].resample("MS").sum()
train, test = monthly.iloc[:-3], monthly.iloc[-3:]
ets = ExponentialSmoothing(train, trend="add", damped_trend=True).fit()
print(ets.forecast(3))
arima = ARIMA(train, order=(1,1,1)).fit()
print(arima.forecast(3))
```

Generated outputs are stored in the Business_Analytics_Flipkart_Project folder.